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### Output Comparison Table

MODEL comparison matrix

| Classification                         | Accuracy | Precision | Recall | F1Score | ROC/AUC |
|----------------------------------------|----------|-----------|--------|---------|---------|
| Logistic Regression                    | 0.884    | 0.812     | 0.845  | 0.828   | 0.931   |
| Decision Tree Classifier               | 0.896    | 0.830     | 0.852  | 0.841   | 0.908   |
| Random Forest Classifier               | 0.942    | 0.905     | 0.918  | 0.911   | 0.976   |
| Gradient Boosting Classifier (XGBoost) | 0.958    | 0.928     | 0.936  | 0.932   | 0.985   |
| K-Nearest Neighbors Classifier         | 0.871    | 0.792     | 0.820  | 0.806   | 0.912   |
| Support Vector Classifier (SVC)        | 0.925    | 0.881     | 0.894  | 0.887   | 0.962   |
| Voting Vs Avearge (best 3 out of 6 )   | 0.961    | 0.931     | 0.940  | 0.935   | 0.988   |
| Ensemble                               | 0.964    | 0.935     | 0.945  | 0.940   | 0.991   |

Which model gave you the best result and which was the worst?

Best: Bayesian Ensemble / XGBoost (ROC-AUC: 0.991, F1: 0.940).

Worst: K-Nearest Neighbors (F1: 0.806)

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